disease but most are vital for life on Earth. Among other things, they break down organic matter so it can be recycled back into the ecosystem.

Trillions of microbes also live on and in the human body. The most common are the beneficial bacteria that produce vitamins, aid food digestion, and help the immune system fight off harmful microbes.

Scientists did not understand microbes until they could see them. The first observations began in the 17th century, using the recently invented microscope. These studies revealed a previously unknown world teeming with microbiotic life. Around the same time, the word “germ,” originally meaning “seed,” was first used to describe these tiny organisms.

“Microbes are the worker bees that perform most of the important functions in your body.”

Dr. Robynne Chutkan,

Microbiome expert and author

Fighting disease

Some 17th and 18th-century scientists believed that certain “germs” might cause diseases, but the prevailing view was that such maladies were the spontaneous result of inherent weakness in an organism. It was not until the painstaking laboratory work of the 19th-century French chemist Louis Pasteur that the “germ theory of disease” was proved.

Pasteur began by looking at the alcohol fermentation process. He discovered that sourness in wine was caused by external agents—microbes, or germs. A crisis in the French silk industry, caused by an epidemic among silkworms, then allowed him to isolate and identify the microorganisms that caused the particular disease. As he extended germ theory to human disease, Pasteur proposed that germs invade the body to cause specific disorders. Edward Jenner, nearly 100 years before, had shown that a disease could be prevented by the use of a “vaccine”—a virus similar to that of the disease-